

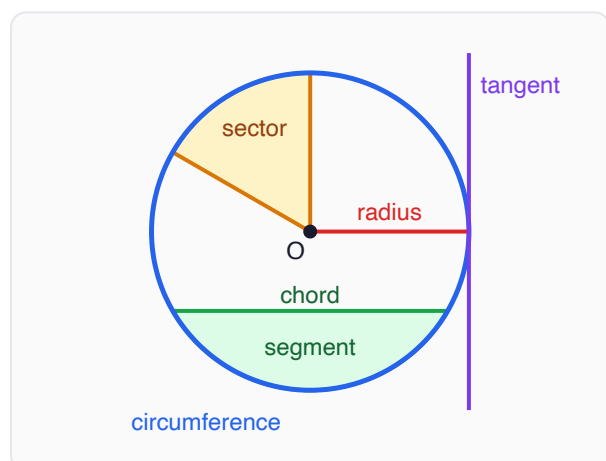
- RAG Learning

4.3 Circles & Circle Theorems

AQA GCSE Mathematics 8300 — Geometry and Measures — spec refs G9, G10, G18 — Papers 1 (non-calculator), 2 and 3 (calculator)

This topic covers the vocabulary of circles, calculating circumference and area, arc lengths and sector areas, and (Higher tier) the standard circle theorems — including how to apply them with reasons and how to prove them.

4.3.1 Parts of a Circle (G9)



Term	Meaning
Centre	The fixed point equidistant from every point on the circle
Radius	Centre to circumference
Diameter	Chord through the centre; $d = 2r$
Circumference	The perimeter of the circle
Chord	Line joining two points on the circle
Tangent	Line touching the circle at one point
Arc	Part of the circumference

Sector	Region bounded by two radii and an arc
Segment	Region bounded by a chord and an arc

Exam Tip: Do not mix up **sector** (two radii, "pizza slice") and **segment** (chord, "cut-off piece"). A chord splits a circle into a *minor* and a *major* segment; two radii split it into a *minor* and a *major* sector. A semicircle is a sector with angle 180° , and a quadrant is a sector with angle 90° .

4.3.2 Circumference and Area

Circumference: $C = \pi d = 2\pi r$

Area: $A = \pi r^2$

Reverse: $r = \frac{C}{2\pi}$ and $r = \sqrt{\frac{A}{\pi}}$

Worked Example: A circle has diameter 7 cm. Find its circumference and area, to 1 decimal place.

$$C = \pi \times 7 = 21.99 \dots \approx \mathbf{22.0 \text{ cm}}$$

$$r = 3.5, \text{ so } A = \pi \times 3.5^2 = 38.48 \dots \approx \mathbf{38.5 \text{ cm}^2}$$

Worked Example: A circle has circumference 12π cm. Find its area in terms of π .

$$2\pi r = 12\pi \Rightarrow r = 6. \quad A = \pi \times 6^2 = \mathbf{36\pi \text{ cm}^2}.$$

On the non-calculator paper you are often asked to leave answers **in terms of π** : treat π like a letter and do not round. For example, a ring between circles of radius 5 cm and 3 cm has area $25\pi - 9\pi = 16\pi \text{ cm}^2$.

Exam Tip: Check whether you have been given the **radius or the diameter**. Using the diameter in πr^2 makes the area four times too big — a very common lost mark.

4.3.3 Arc Length and Sector Area (G18)

A sector with angle θ is the fraction $\frac{\theta}{360}$ of the whole circle, so its arc is that fraction of the circumference and its area is that fraction of the circle's area.

$$\text{Arc length} = \frac{\theta}{360} \times 2\pi r$$

$$\text{Sector area} = \frac{\theta}{360} \times \pi r^2$$

$$\text{Perimeter of a sector} = \text{arc length} + 2r$$

Worked Example: A sector has radius 6 cm and angle 120° . Find (a) the arc length, (b) the area, (c) the perimeter.

$$(a) \frac{120}{360} \times 2\pi \times 6 = \frac{1}{3} \times 12\pi = 4\pi \approx \mathbf{12.6 \text{ cm}}$$

$$(b) \frac{120}{360} \times \pi \times 6^2 = \frac{1}{3} \times 36\pi = 12\pi \approx \mathbf{37.7 \text{ cm}^2}$$

$$(c) 4\pi + 6 + 6 = 4\pi + 12 \approx \mathbf{24.6 \text{ cm}}$$

Working backwards to find the angle or radius

Worked Example: A sector has radius 5 cm and arc length 7 cm. Find the angle.

$$\frac{\theta}{360} \times 2\pi \times 5 = 7 \Rightarrow \theta = \frac{7 \times 360}{10\pi} = 80.21 \dots \approx \mathbf{80.2^\circ}$$

Worked Example: A sector has angle 40° and arc length 8π cm. Find the radius.

$$\frac{40}{360} \times 2\pi r = 8\pi \Rightarrow \frac{2\pi r}{9} = 8\pi \Rightarrow 2r = 72 \Rightarrow r = \mathbf{36 \text{ cm}}$$

Area of a segment

Segment area = sector area — area of the triangle formed by the two radii and the chord.

Worked Example: Radius 10 cm, angle at centre 90° . Find the area of the minor segment.

$$\text{Sector} = \frac{90}{360} \times \pi \times 10^2 = 25\pi. \text{ Triangle} = \frac{1}{2} \times 10 \times 10 = 50.$$

$$\text{Segment} = 25\pi - 50 = 28.54 \dots \approx \mathbf{28.5 \text{ cm}^2}$$

Worked Example: Higher Radius 8 cm, angle at centre 60° . Find the area of the minor segment.

$$\text{Sector} = \frac{60}{360} \times \pi \times 64 = 33.51 \dots \text{ Triangle} = \frac{1}{2} \times 8 \times 8 \times \sin 60^\circ = 27.71 \dots$$

$$\text{Segment} = 33.51 - 27.71 \approx \mathbf{5.80 \text{ cm}^2}$$

Composite shapes

Worked Example: A window is a rectangle 1.2 m wide and 1.5 m tall with a semicircle on top. Find its perimeter.

$$\text{Curved edge} = \frac{1}{2} \times \pi \times 1.2 = 0.6\pi = 1.885 \text{ m. Straight edges} = 1.2 + 1.5 + 1.5 = 4.2 \text{ m.}$$

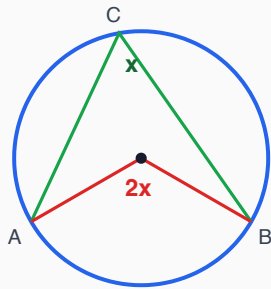
$$\text{Perimeter} = 4.2 + 1.885 = \mathbf{6.09 \text{ m}}$$
 (the top of the rectangle is inside the shape, so it is not counted).

Exam Tip: For a perimeter, trace round the outside edge with your finger and list every piece. The straight sides of a sector (two radii) and the diameter of a semicircle are easy to forget.

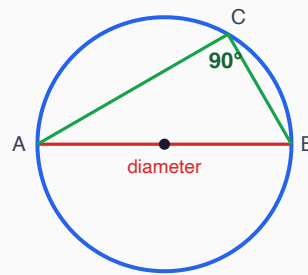
4.3.4 The Circle Theorems (G10) Higher

In AQA GCSE, the circle theorems are Higher-tier content. You must be able to **apply** them, giving the theorem as a reason, and **prove** them. Learn the exact wording — "circle theorem" on its own scores nothing.

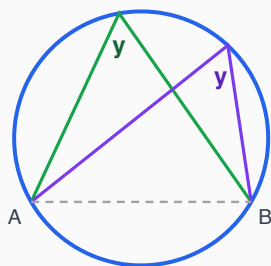
#	Theorem (wording to use as a reason)
1	The angle at the centre is twice the angle at the circumference (subtended by the same arc).
2	The angle in a semicircle is 90° .
3	Angles in the same segment are equal.
4	Opposite angles of a cyclic quadrilateral add up to 180° .
5	The tangent at any point is perpendicular to the radius at that point.
6	Tangents from an external point are equal in length.
7	The perpendicular from the centre to a chord bisects the chord.
8	Alternate segment theorem: the angle between a tangent and a chord equals the angle in the alternate segment.



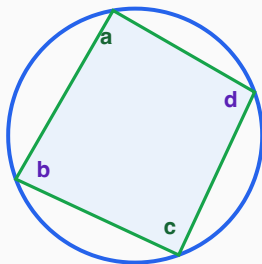
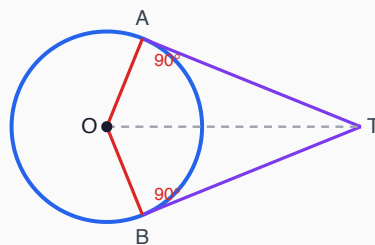
1. Centre = 2 × circumference

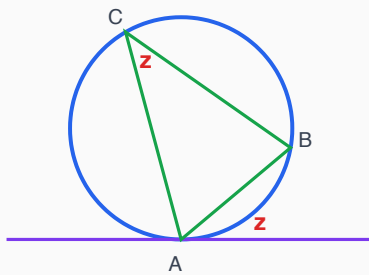


2. Angle in a semicircle



3. Angles in the same segment

4. $a + c = 180^\circ$, $b + d = 180^\circ$ 5 & 6. $OA \perp TA$ and $TA = TB$



8. Alternate segment theorem

A **cyclic quadrilateral** is a quadrilateral whose four vertices all lie on the circumference of a circle. The **alternate segment** is the segment on the other side of the chord from the angle between the tangent and the chord.

Applying the theorems

Worked Example: A and B lie on a circle centre O and angle $OAB = 28^\circ$. C is on the major arc AB. Find angle ACB .

$OA = OB$ (radii), so triangle OAB is isosceles and angle $OBA = 28^\circ$.

Angle $AOB = 180 - 28 - 28 = 124^\circ$ (angles in a triangle).

Angle $ACB = 124 \div 2 = 62^\circ$ (angle at the centre is twice the angle at the circumference).

Worked Example: TA and TB are tangents touching a circle centre O at A and B. Angle $ATB = 40^\circ$. Find angle AOB and angle OAB .

Angles $OAT = OBT = 90^\circ$ (tangent is perpendicular to the radius).

Angle $AOB = 360 - 90 - 90 - 40 = 140^\circ$ (angles in a quadrilateral).

Triangle OAB is isosceles ($OA = OB$), so angle $OAB = (180 - 140) \div 2 = 20^\circ$.

Worked Example: ABCD is a cyclic quadrilateral with angle $ABC = 3x$ and angle $ADC = x + 40^\circ$. Find x .

$3x + x + 40 = 180$ (opposite angles of a cyclic quadrilateral add up to 180°)
 $4x = 140$, so $x = 35$, giving angle $ABC = 105^\circ$ and angle $ADC = 75^\circ$.

Worked Example: A chord of length 24 cm is in a circle of radius 13 cm. How far is the chord from the centre?

The perpendicular from the centre bisects the chord, so the half-chord is 12 cm.

Distance = $\sqrt{13^2 - 12^2} = \sqrt{25} = 5$ cm.

Worked Example: The tangent at P makes an angle of 58° with chord PQ. R is in the alternate segment and angle $PQR = 71^\circ$. Find angle QPR .

Angle $PRQ = 58^\circ$ (alternate segment theorem).

Angle $QPR = 180 - 58 - 71 = 51^\circ$ (angles in a triangle).

Exam Tip: Look for hidden **isosceles triangles**: any triangle with two radii as sides is isosceles. Mark radii and tangents on the diagram first, then add every angle you can find, writing a reason beside each one.

4.3.5 Proving the Circle Theorems Higher

AQA expects you to prove the standard theorems and to use them to prove related results. Most proofs start by drawing in radii to create isosceles triangles.

Worked Example: Prove that the angle at the centre is twice the angle at the circumference.

Join C to O and extend the line to a point D. $OA = OC$ (radii), so triangle OAC is isosceles: let angle $OAC = \text{angle } OCA = x$.

The exterior angle of a triangle equals the sum of the interior opposite angles, so angle

$$AOD = 2x.$$

Similarly $OB = OC$: let angle $OBC = \text{angle } OCB = y$, so angle $BOD = 2y$.

Angle $ACB = x + y$ and angle $AOB = 2x + 2y = 2(x + y)$. Hence the angle at the centre is twice the angle at the circumference. ■

Worked Example: Prove that the angle in a semicircle is 90° .

AB is a diameter with centre O; C is on the circle. $OA = OB = OC$ (radii), so triangles OAC and OBC are isosceles.

Let angle $OAC = \text{angle } OCA = x$ and angle $OBC = \text{angle } OCB = y$.

Angles in triangle ABC: $x + y + (x + y) = 180$, so $2(x + y) = 180$ and $x + y = 90$.

Angle $ACB = x + y = 90^\circ$. ■ (Alternatively: angle $AOB = 180^\circ$, so by the centre theorem angle $ACB = 90^\circ$.)

Worked Example: Prove that opposite angles of a cyclic quadrilateral add up to 180° .

Let ABCD be cyclic with centre O, angle $BAD = x$ and angle $BCD = y$. Join OB and OD. The angle at O on the side of C is $2x$, and the reflex angle at O on the side of A is $2y$ (centre theorem).

Together they make a full turn: $2x + 2y = 360$, so $x + y = 180^\circ$. ■

Worked Example: Prove that the perpendicular from the centre to a chord bisects the chord.

Let M be the foot of the perpendicular from O to chord AB. In triangles OMA and OMB: angle $OMA = \text{angle } OMB = 90^\circ$; $OA = OB$ (radii, the hypotenuses); OM is common.

So the triangles are congruent (RHS) and $AM = MB$. ■

Worked Example: Prove that tangents from an external point are equal.

Tangents from T touch the circle at A and B. Angle $OAT = \text{angle } OBT = 90^\circ$ (tangent

perpendicular to radius); $OA = OB$ (radii); OT is common.

So triangles OAT and OBT are congruent (RHS), and $TA = TB$. ■

Worked Example: Outline proof of the alternate segment theorem.

Let the tangent at A make angle z with chord AB. Draw the diameter AD; angle $DAT = 90^\circ$ (tangent perpendicular to radius), so angle $DAB = 90 - z$.

Angle $ABD = 90^\circ$ (angle in a semicircle), so angle $ADB = 180 - 90 - (90 - z) = z$.

Any angle ACB in the same segment as D equals angle ADB (angles in the same segment), so angle $ACB = z$. ■

Exam Tip: In a "prove" question, every step needs a reason, and you must reach the required statement exactly. Use letters for unknown angles (x , y) rather than specific numbers — a numerical example is not a proof.

Summary

Topic	Key facts
Circumference and area	$C = \pi d = 2\pi r$, $A = \pi r^2$
Arc length	$\frac{\theta}{360} \times 2\pi r$
Sector area	$\frac{\theta}{360} \times \pi r^2$
Segment area	Sector — triangle
Circle theorems Higher	Centre = $2 \times$ circumference; semicircle 90° ; same segment equal; cyclic quad 180° ; tangent $\perp$ radius; tangents equal; perpendicular bisects chord; alternate segment

